MAIS PAC

IPhO 04.04: Approximation, Taylor Series and Numerical Methods · Lecture + Problems

Approximation, Taylor Series and Numerical Methods

MAIS Physics & Astronomy Club (PAC) · IPhO lecture + problem handout

6 lecture sections · 10 problems · 1 interactive experiment · 5 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to olympiad measurements. Five vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice. The coverage checklist is the IPhO syllabus, not this handout.

Contents & learning path

1. Why approximation is the core physics skill
2. Small quantities and Taylor series
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1. Why approximation is the core physics skill

An exact answer to an approximate model is worth less than an approximate answer to the right model. Physics papers are full of situations where the exact equation cannot be solved, and the examiner is testing whether you can recognise the regime in which it becomes solvable, solve it there, and then say how much you gave away.

Approximation appears in three guises, all of which earn marks:

- **Linearisation** for small displacements, small angles, small currents.
- **Series truncation**, where you keep the first term and quote the second as an error.
- **Order-of-magnitude estimation**, where the exponent is the answer and one significant figure is generous.

2. Small quantities and Taylor series

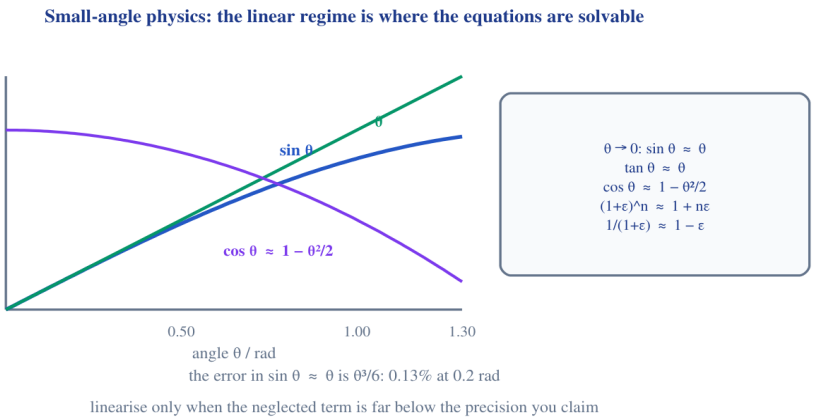


Figure 1. Approximation is not sloppiness: it is the deliberate discarding of a term you can bound. The bound is what makes the answer defensible, and quoting it is what earns the mark.

For a smooth function expanded about zero,

$$f(x) = f(0) + f'(0)x + (f''(0))/(2)x^2 + \quad (1)$$

which gives the working set of approximations, valid for small ε:

$$\sin \varepsilon \approx \varepsilon,\ \cos \varepsilon \approx 1-(\varepsilon^2)/(2),\ e\text{^}\varepsilon \approx 1+\varepsilon,\ (1+\varepsilon)\text{^}n \approx 1+n\varepsilon,\ (1)/(1+\varepsilon) \approx 1-\varepsilon. \quad (2)$$

The last two are the same statement in disguise, and they solve most "small correction" questions. A reliable procedure:

1. Write the exact relation.
2. Identify the small quantity ε (a ratio of lengths, an angle in radians, a resistance ratio).
3. Expand, keeping enough terms that the first neglected one is smaller than the precision asked for.
4. State the size of the neglected term — this is what converts an approximation into a defensible result.

For example, a pendulum's period grows as T = T_0(1+θ_0²/16); at θ_0 = 0.1 {rad} the correction is 6×10^(-4), so a school pendulum timed to 0.1% does not notice it.

3. Graphical and algebraic approximation

Two further techniques are worth naming:

- **The tangent-line (linear) approximation:** near a point, f(x)≈ f(x_0) + f'(x_0)(x-x_0). This is what "the gradient is constant over a small range" means, and it justifies treating any smooth curve as a straight line locally.
- **The binomial expansion of a physical formula:** for r ≫ a, (1)/((r+a)²) = (1)/(r²)(1+(a)/(r))^(-2) ≈ (1)/(r²)(1-(2a)/(r)). This is exactly the multipole expansion, and the same step produces the dipole field, the tidal force, and the small correction to g with altitude.

4. Numerical methods

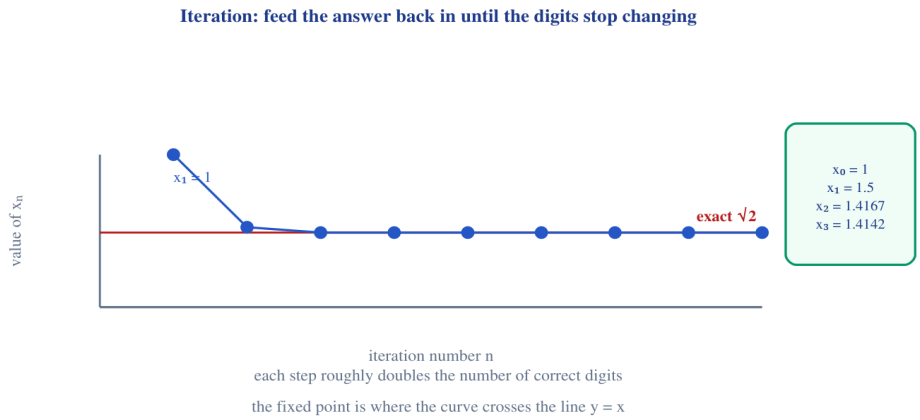


Figure 2. Numerical methods replace an algebraic solution by a sequence of better guesses. The two questions to ask are always: does it converge, and how fast — because a method that converges is useless if it needs a million steps.

Some problems have no closed-form solution and must be iterated. Three methods cover what an olympiad asks:

- **Fixed-point iteration:** rearrange f(x)=0 as x = g(x) and iterate. It converges if |g'(x)|<1 near the root; the error shrinks by that factor each step.
- **Newton–Raphson:** x_(n+1) = x_n - (f(x_n))/(f'(x_n)). Converges quadratically — the number of correct digits roughly doubles per step — which is why it is the default.
- **Bisection:** bracket a root, halve the interval, keep the half where the sign changes. Slow but unstoppable, and it never diverges.

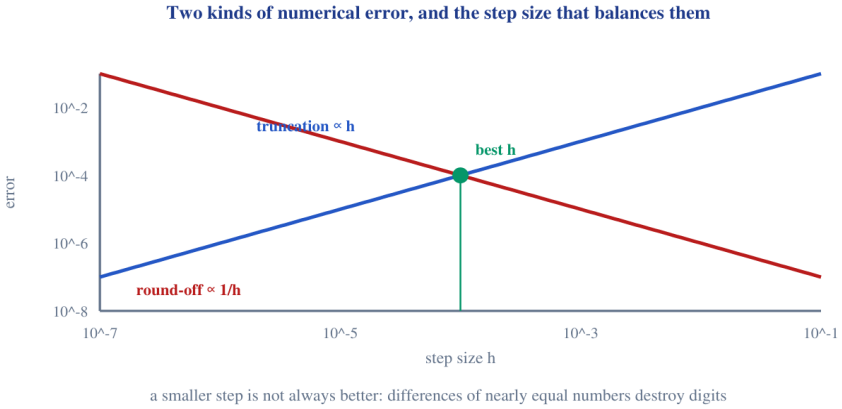


Figure 3. Every numerical estimate trades a truncation error, which falls with a finer step, against a round-off error, which grows as the step shrinks and differences of nearly equal numbers lose significant digits. The crossing point is the best step.

There is a trap that examiners like: a finer step is not always more accurate. Truncation error falls as the step shrinks, but round-off error grows, because subtracting two nearly equal numbers destroys significant digits. The two curves cross at an optimum step, typically around 10^(-7)–10^(-8) for double precision. Quoting that trade-off is worth a mark.

5. Checking an answer

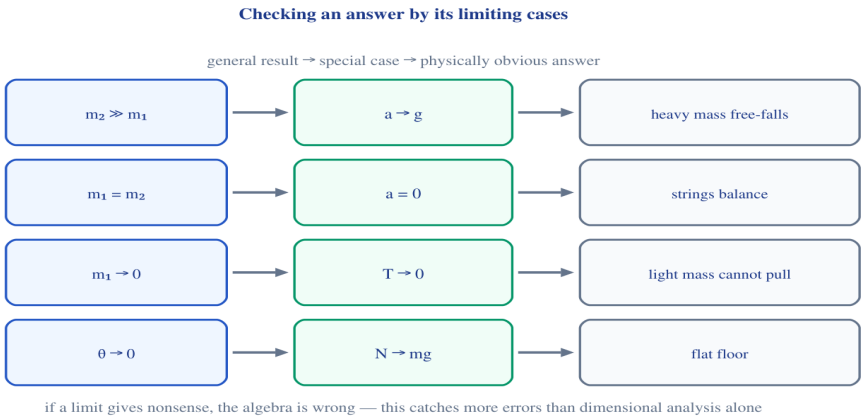


Figure 4. A general formula must reproduce the obvious special cases. Checking two or three limits takes twenty seconds and catches most sign errors and inverted factors.

Three checks catch nearly every algebraic slip, and each takes seconds:

1. **Dimensions** of every term.
2. **Limiting cases:** what happens when a mass is huge, a resistance is zero, an angle is 90°? The answer should reduce to something obvious.
3. **Symmetry:** if the setup is symmetric, the answer must be too. A formula that is not symmetric in two identical masses is wrong.

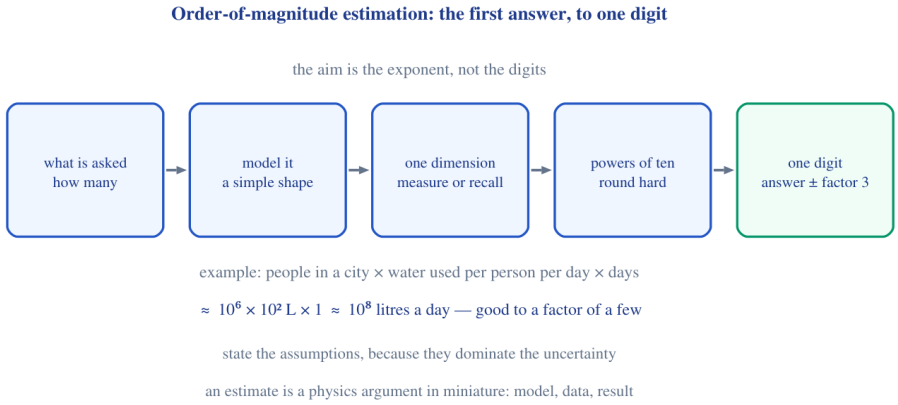


Figure 5. Fermi estimation is examined because it reveals physical intuition. The grader wants the model, the numbers and the exponent — and a comment on which assumption dominates the error.

An order-of-magnitude estimate is a physics argument in miniature: model the system with one simple shape, choose one number for each quantity, work in powers of ten, and quote the result to one digit with its dominant assumption named.

Interactive experiment: How good is the small-angle approximation? — open the online handout to use it.

Approximation etiquette. Write the approximation with its condition, not just the formula: "for $\theta \ll 1$, $\sin\theta \approx \theta$" rather than "$\sin\theta = \theta$". Then quote the size of the first neglected term. An answer that says "$\theta \approx 0.2$ rad, so the error in $\sin\theta \approx \theta$ is $\theta^3/6 \approx 1.3 \times 10^{-3}$, below the 1% precision of the data" is worth full marks; the same number without the justification is worth half.

6. Problems: reason, calculate, check

Ten original problems in three tiers: **[1]** foundation, **[2]** multi-step olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. State assumptions, keep units, and estimate the answer before computing it. Every problem has a worked solution with a consistency check.

[1] Problem 1 — Small-angle estimates.

(a) Estimate sin (3.0°) using the small-angle approximation. (b) Find the percentage error compared with the exact value. (c) At what angle does the approximation reach 1% error?

Worked solution →

[1] Problem 2 — The binomial approximation. A satellite is at height h above the Earth's surface, with h ≪ R. (a) Write g at that height. (b) Use the binomial approximation to show that the fractional decrease is about 2h/R. (c) Evaluate for h = 400 {km}.

Worked solution →

[2] Problem 3 — Which term dominates? For a pendulum, the exact equation is θ = -(g)/(L)sin θ. (a) Expand sin θ to two terms. (b) Show that the first correction to the period is of order θ_0². (c) Estimate the amplitude at which the period increases by 1%.

Worked solution →

[2] Problem 4 — Newton–Raphson by hand.

(a) Set up Newton–Raphson for √(10). (b) Starting from x_0 = 3, carry out two iterations. (c) Explain why each iteration roughly doubles the number of correct digits.

Worked solution →

[2] Problem 5 — Bisection.

(a) Show that x³ - x - 1 = 0 has a root between 1 and 2. (b) Carry out three bisection steps. (c) State how many steps would give six decimal places, and why bisection is used despite its slowness.

Worked solution →

[2] Problem 6 — Round-off and catastrophic cancellation. A student computes √(1.0001)-1 in single precision (about 7 significant digits). (a) Explain why the result loses accuracy. (b) Use the binomial approximation to compute the same quantity analytically. (c) State the general rule for avoiding the problem.

Worked solution →

[2] Problem 7 — A Fermi estimate. Estimate the mass of air in a room 5 {m}×4 {m}×3 {m} at atmospheric pressure. (a) State the model and the numbers you use. (b) Carry out the estimate to one significant figure. (c) State which assumption dominates the uncertainty.

Worked solution →

[3] Problem 8 — A small correction with a sign. A simple pendulum of length L hangs in a lift accelerating upwards at a. (a) Find the effective gravity. (b) Show that for a≪ g the fractional change of period is about -a/2g. (c) For a = 0.10g, compare the approximation with the exact value and comment on the sign.

Worked solution →

[3] Problem 9 — Iterating a transcendental equation. The current in a diode circuit satisfies I = I_s(e^(V/V_T)-1) with I_s = 10^(-12) {A} and V_T = 26 {mV}, in series with a resistor R = 1.0 {kΩ} and a 1.5 {V} cell. (a) Write the equation that determines V. (b) Explain why it cannot be solved algebraically. (c) Estimate the operating point with two approximations and say which regime the diode is in.

Worked solution →

[3] Problem 10 — Choosing a method. You must solve x = cos x. (a) Show that a solution exists in (0,1). (b) Explain why fixed-point iteration converges here, and find the fixed point to three decimal places. (c) Compare the convergence rate with Newton–Raphson and say which you would use in an exam.

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this IPhO syllabus topic. They are not copied from official papers and are not official marking schemes.

Sources and scope. Topic coverage follows the IPhO syllabus (accepted 2014, amended 2015) as published by the IPhO organisation. Official problems are available through the IPhO past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.

Where to learn more

IPhO syllabus and statutes

> The coverage checklist this curriculum follows: accepted 2014, amended 2015. Read it once for scope, not for depth.
>
> IPhO past problems and solutions
>
> The only honest test of readiness. Work recent theoretical and experimental papers under time, then compare with the official marking schemes.
>
> Paul's Online Math Notes
>
> Calculus, series and approximation practice at exactly the level the syllabus expects.